Ten-day river forecasts for the Tennessee basin: the graph model against the LSTM and the National Water Model

Results report, 27 October 2026: at gauges, at gauges the model never saw, at all 58,834 river reaches, for floods, in basins without dams, and against TVA's own forecasts.

Summary

We forecast river flow every six hours for the next ten days at all 58,834 NHDPlus reaches of the Tennessee River basin. Our model, called **GNN** here, is a graph neural network that runs an LSTM at every reach, mixes neighbouring reaches through graph layers, routes water downstream while conserving mass, and forecasts each gauge directly from its recent flow. It is compared with two baselines: a tuned **LSTM** (the standard machine-learning model in hydrology) and the **National Water Model (NWM)** operational forecast.

All models follow one forecast rule: observed weather and flow up to the issue time, and after it only the GEFSv12 rain forecast. The test covers 1,917 daily forecasts issued from 2020-09-23 to 2025-12-22, years never used for training or model selection. Headline scores (median NSE over days 1-5, with 95% intervals; NSE is 1 for a perfect forecast and 0 for one no better than the observed mean):

Test	GNN	LSTM	NWM
1. Gauged: recent flow shown (101 gauges)	0.415 (0.39 to 0.46)	0.414 (0.38 to 0.44)	0.253 (0.20 to 0.30)
2. Ungauged: gauges never seen (98 gauges)	**0.378 (0.34 to 0.41)**	0.111 (0.04 to 0.18)	0.258 (0.20 to 0.30)
3. Every reach, vs NWM retrospective (52,126 reaches)	**0.057**	–7.57	–0.80

In one sentence: the GNN ties the LSTM where a gauge's recent flow is available, is clearly the best model where it is not (gauges it never saw and the whole river network), and beats NWM in all three tests over days 1-5; like both baselines, it badly underpredicts the largest floods.

Main findings

- **Gauged:** a tie (0.415 against 0.414); the LSTM is better in days 1-2, the GNN from day 3 to day 10. The GNN beats NWM at 90% of gauges.
- **Ungauged:** 0.378 against 0.111 (LSTM) and 0.258 (NWM). Losing the gauge costs the GNN 0.03 NSE, the LSTM 0.29.
- **Headwater basins without dams** (54 gauges): the GNN is ahead at gauges (0.399 against 0.371 and 0.157; intervals overlap with the LSTM) and clearly ahead without them (0.353 against 0.088 and 0.157).
- **TVA's own forecasts** (29 gauges): TVA is best at 1-3 days (72 h: 0.446 against 0.355), the GNN from day 4 (120 h: 0.310 against 0.170). TVA's own rain closes about half of TVA's early lead.
- **Every reach:** the GNN is closer to the NWM retrospective than the NWM forecast at 65% of reaches.
- **Floods:** every model's peaks above the 99th percentile are 56-80% too low; the GNN's 99th-percentile forecast is about right (+21% one day ahead). Helene was missed by every model.

1. What is being compared

The Tennessee River basin (about 106,000 km^2, parts of seven states) is represented by the NHDPlus river network: 58,834 reaches, each a short river segment with its own small catchment. Only 128 reaches have a public flow gauge, so more than 99.8% of the network is ungauged. Many rivers are regulated by TVA dams.

Tennessee River basin: 58,834 modelled reaches, 128 observed-flow gauges

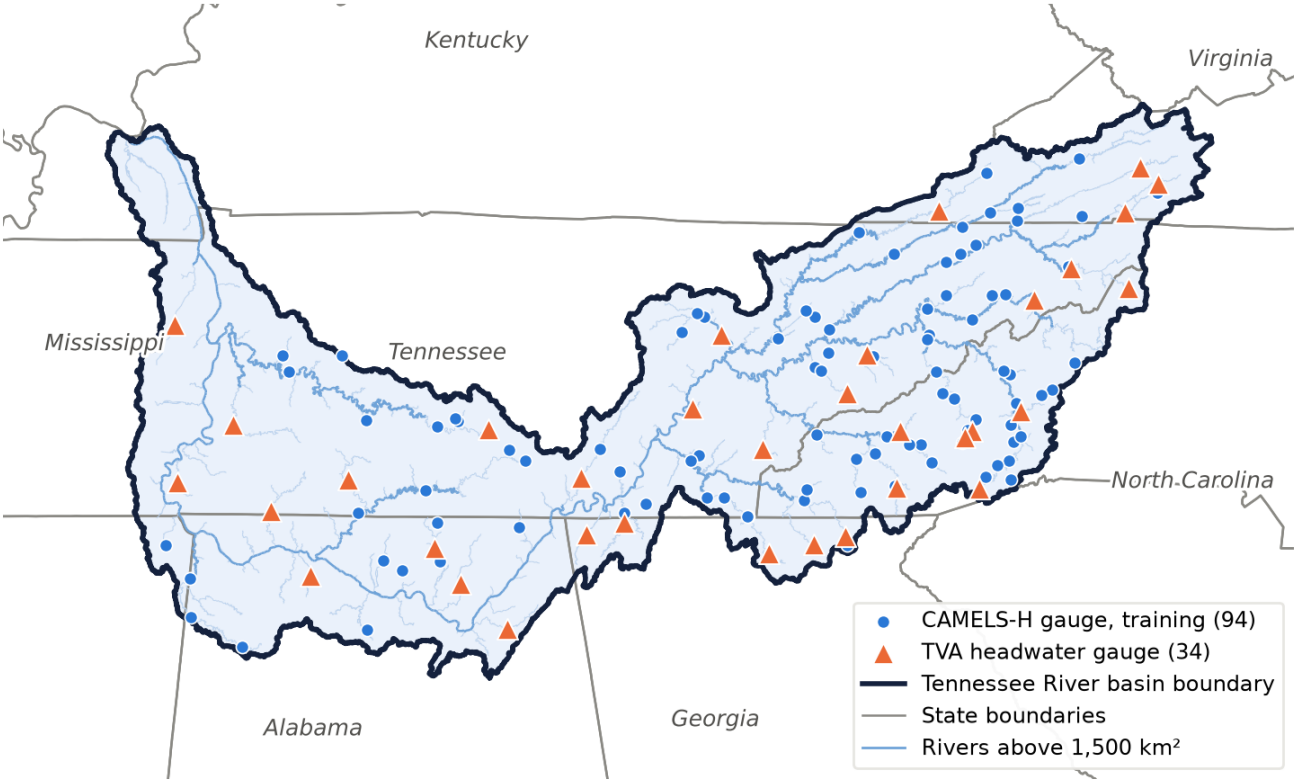


Figure 1. The basin, its main rivers and the 128 observed-flow gauges. The GNN forecasts every one of the 58,834 reaches, not only the gauges.

1.1 The forecast rule (the same for every model)

A forecast is issued at 00 UTC each day and covers the next 40 six-hour steps (10 days). Before the issue time a model may use observed AORC weather and observed USGS flow; after it, only the GEFSv12 control-member rain forecast to 240 h. Training uses the GEFSv12 reforecast (2000-2018), model selection 2019, and the test the operational GEFSv12 from 23 September 2020, its first day. Forecasts made with the rain that actually fell are reported only as a "realized rain" diagnostic: an upper bound for a perfect rain forecast.

1.2 The GNN

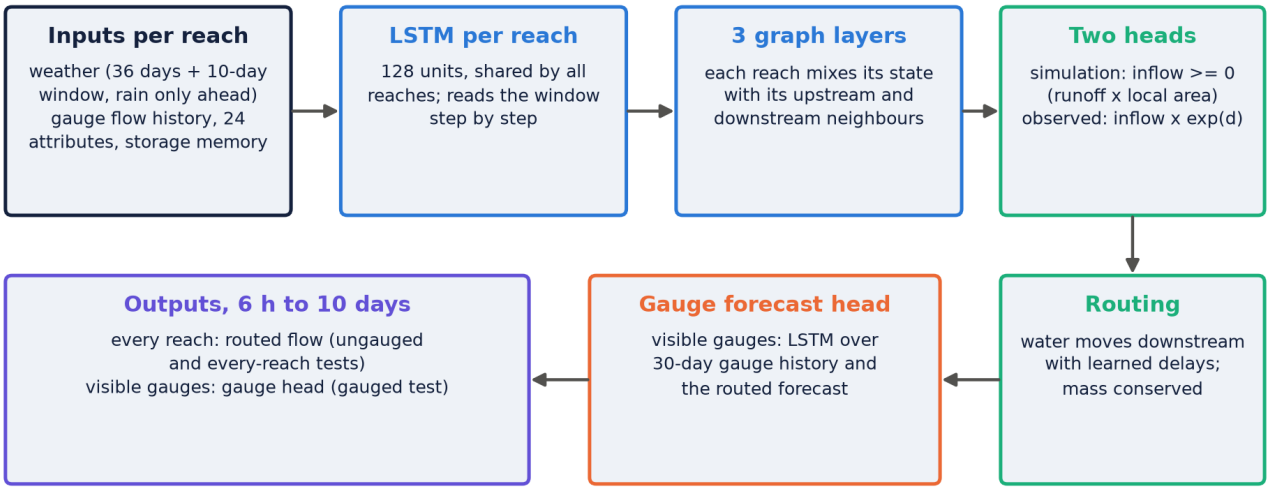


Figure 2. The GNN. Every part is shared by all reaches; one forward pass forecasts the whole network. About 930,000 parameters.

- **LSTM per reach** (128 units): reads each reach's weather, season, catchment attributes, antecedent storage (weighted rain over 7-365 days) and, at a visible gauge, its observed flow, over 36 days of history and the 10-day forecast window.
- **Graph layers** (3): each reach mixes its state with the mean state of its upstream and downstream neighbours at every step, so nearby reaches and gauges share information.
- **Routing**: each reach's runoff is moved downstream with learned travel-time curves that each sum to one, so water is conserved and flow grows with drainage area by construction.
- **Gauge head** (an LSTM with 256 units): at a gauge whose recent flow is shown, it reads 30 days of observed flow, the routed forecast, basin-mean forecast rain and basin attributes, and forecasts the gauge directly.
- **Tail head**: also gives the 50th, 90th and 99th percentiles of the flow at gauges, for flood warnings.

Training. Stage A trains the network on the NWM v3.0 retrospective at every reach (no gauge data). Stage B trains the whole model jointly on observed flow at the gauges: an NSE loss on the forecasts, a quantile (pinball) loss on the tail head, and a prior that keeps every reach close to NWM (weight 0.3). A random half of the gauges show their recent flow in each sample (none in 15%), so one model learns both the gauged and the ungauged case. For the ungauged test, stage B is repeated five times, each time withholding one fifth of the gauges completely.

1.3 The baselines

- **LSTM, tuned.** One lumped encoder-decoder LSTM (256 units) shared by all gauge basins: basin-mean weather and attributes, 30 days of the gauge's flow, and GEFS rain to 240 h in; a 10-day forecast out. Settings tuned on 2019. Like the GNN, it is trained once with all gauges and five times with one fold of gauges withheld. For the every-reach test it is run at each reach with that reach's upstream-basin means (no flow history).
- **NWM forecast.** NOAA's physics-based operational medium-range forecast at every reach. It assimilates USGS observations, which favours it at the first hours, and may use the held-out gauges, so it is a reference rather than a fair competitor in the ungauged test.

2. How the models are tested

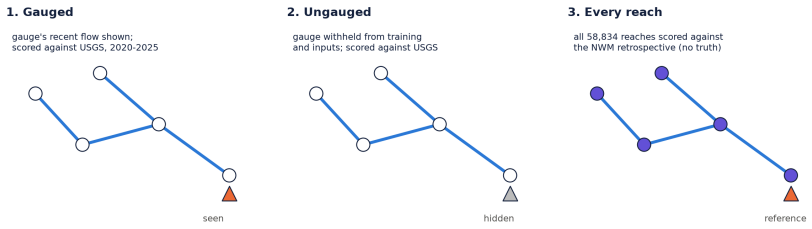


Figure 3. The three tests.

Test	Truth	Where	When	Question
1. Gauged	USGS observed flow	101 gauges, recent flow shown	1,917 issues, 2020-2025	How good is the forecast where a gauge exists?
2. Ungauged	USGS observed flow	98 gauges, each held out in one of 5 folds	same issues	How good is it where no gauge exists?
3. Every reach	NWM v3.0 retrospective (no observations exist)	all 58,834 reaches	425 issues, 2020-09 to 2023-01 (end of the retrospective)	Is the whole network realistic?
Headwater basins (section 8)	as tests 1-3	54 gauges with no dam or gauge upstream; reaches with no dam upstream	as tests 1-3	How do the models compare on natural flow?
TVA forecasts (section 9)	USGS / TVA observed flow	29 TVA headwater gauges, recent flow shown	same issues	How does the GNN compare with TVA's operational forecast?

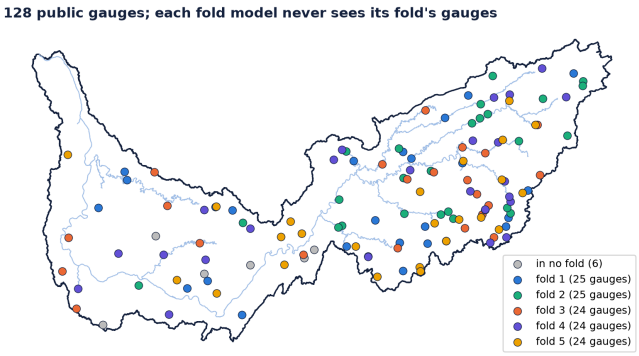


Figure 4. The gauges and their folds. In the ungauged test every gauge is forecast by the fold model that never saw its flow (no targets, inputs or model selection).

Scoring

- Every model is scored on the same issues, gauges, observations and missing-value masks, with the same GEFS rain.
- NSE is computed per gauge (or reach) and lead time over all issues. **Days 1-5** is each gauge's mean NSE over the 24, 48, 72, 96 and 120 h leads; we report the median over gauges with a 95% bootstrap interval (resampling gauges).
- Flood skill uses each gauge's 90th and 99th percentile flow in the training years as thresholds: the critical success index (CSI), hit rate and false-alarm ratio, and the error in the size of observed peaks above the 99th percentile.

3. Results at a glance

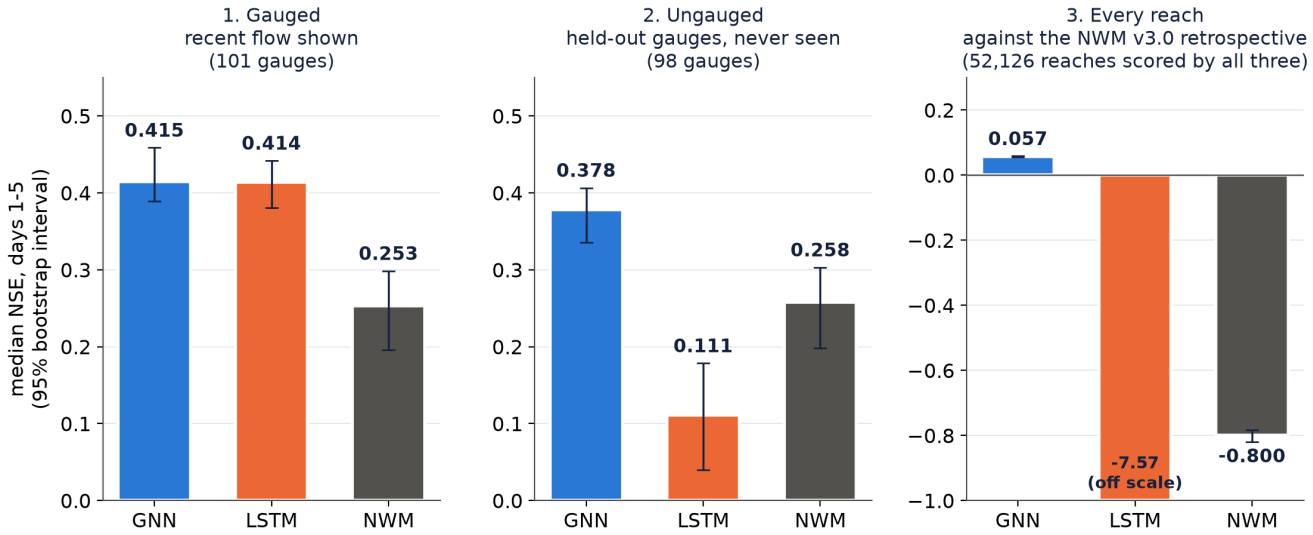


Figure 5. The three tests side by side: median NSE over days 1-5 with 95% bootstrap intervals. Every reach: reaches scored by all three models; the LSTM's median is far below the axis.

The GNN is the only model that is good in all three tests. At gauges it ties the tuned LSTM (the intervals overlap almost completely) and is well ahead of NWM. At gauges it never saw it keeps most of its skill (0.378), while the LSTM falls to 0.111, below NWM. Across the network it is the only model with a positive median NSE against the retrospective.

4. Gauged and ungauged gauges

4.1 Skill by lead time

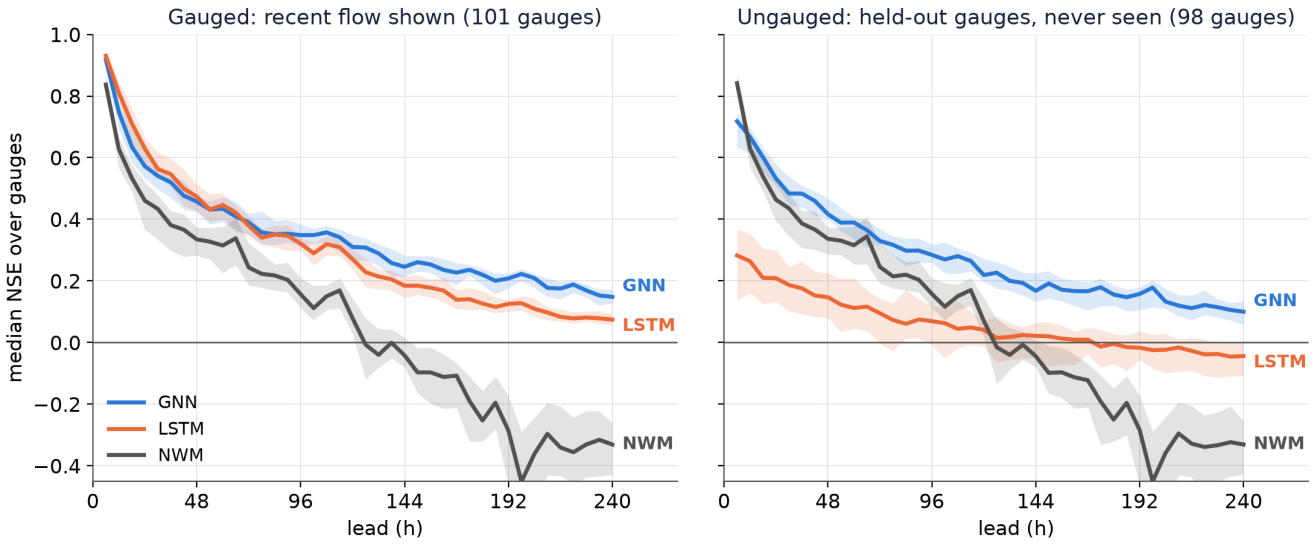


Figure 6. Median NSE over gauges at every lead from 6 h to 10 days (bands: 95% intervals). Left: the recent flow of the gauge is shown to the model. Right: each gauge is forecast by the fold model that never saw it.

Gauged, median NSE	6 h	24 h	72 h	120 h	240 h	Days 1-5
GNN	0.918	0.572	0.390	0.310	0.148	0.415
LSTM	0.931	0.629	0.379	0.270	0.074	0.414
NWM forecast	0.837	0.461	0.244	0.079	–0.331	0.253
Persistence (last observed flow)	0.790	0.085	–0.415	–0.571	–0.687	–0.327
GNN, realized rain (diagnostic)	0.920	0.744	0.699	0.599	0.368	0.677
LSTM, realized rain (diagnostic)	0.930	0.712	0.643	0.536	0.304	0.634

Ungauged, median NSE	6 h	24 h	72 h	120 h	240 h	Days 1-5
GNN	0.718	0.532	0.330	0.219	0.100	0.378
LSTM	0.283	0.209	0.095	0.041	–0.044	0.111
NWM forecast	0.842	0.463	0.245	0.070	–0.331	0.258
GNN, realized rain (diagnostic)	0.713	0.687	0.617	0.520	0.320	0.623
LSTM, realized rain (diagnostic)	0.283	0.297	0.232	0.177	0.058	0.216

- **Gauged:** the LSTM leads in the first two days, the curves cross near 60 h, and the GNN leads to 10 days. Both machine-learning models beat NWM and persistence at every lead.
- **Ungauged:** the GNN leads the LSTM at every lead, by 0.43 NSE at 6 h and 0.15 at 10 days. NWM is better at 6 h only, where it benefits from assimilating observations.
- **Rain is the main limit.** With the rain that fell, days 1-5 rise to 0.677 (gauged) and 0.623 (ungauged) for the GNN. The GNN gains more from better rain than the LSTM does.

4.2 Gauge by gauge

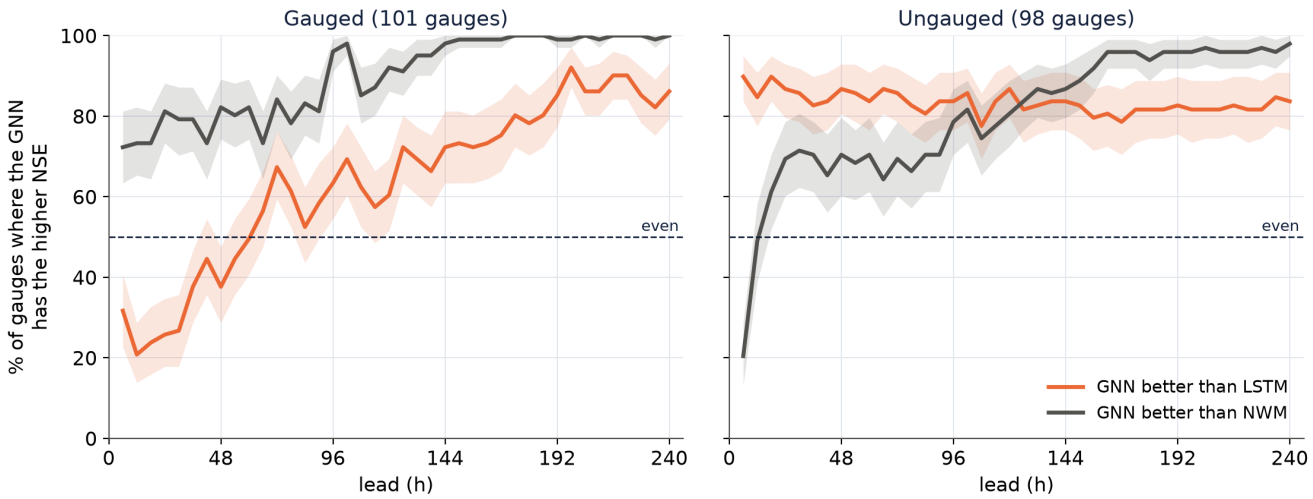


Figure 7. Share of gauges where the GNN has the higher NSE than each baseline, by lead (bands: 95% intervals; dashed: an even split).

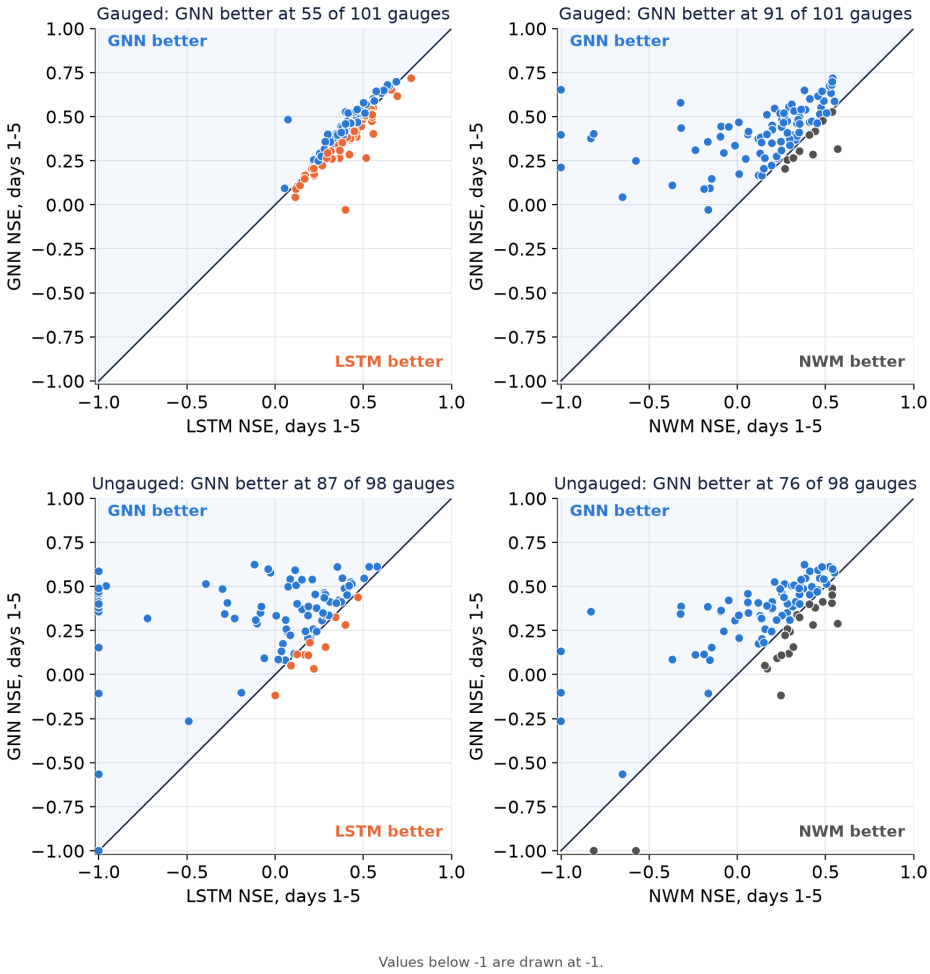


Figure 8. Days 1-5 NSE at each gauge: the GNN (vertical) against each baseline (horizontal). Points above the diagonal are gauges where the GNN is better.

Gauged, the GNN and the LSTM are close at most gauges (the points hug the diagonal) and the GNN is better at 54% of them. Ungauged, the LSTM collapses at many gauges, often below NSE 0, while the GNN stays positive at almost all of them.

4.3 What the gauge's own flow is worth

The same held-out gauges (days 1-5, median NSE)	Value
GNN, recent flow shown	0.412
GNN, all-gauge model, recent flow hidden	0.421
GNN, fold model that never saw the gauge	0.378
LSTM, recent flow shown	0.416
LSTM, fold model that never saw the gauge	0.111
NWM forecast	0.258

The LSTM depends on the gauge's history: without it, it loses about 0.3 NSE. The GNN loses little, because the network, routing and NWM prior carry most of the skill. The flip side is that showing the gauge's flow does not help the GNN over days 1-5 (it helps only at the shortest leads), which is why it does not pull ahead of the LSTM at gauges.

5. Where the GNN wins and where it loses

5.1 By gauge type (TVA dams)

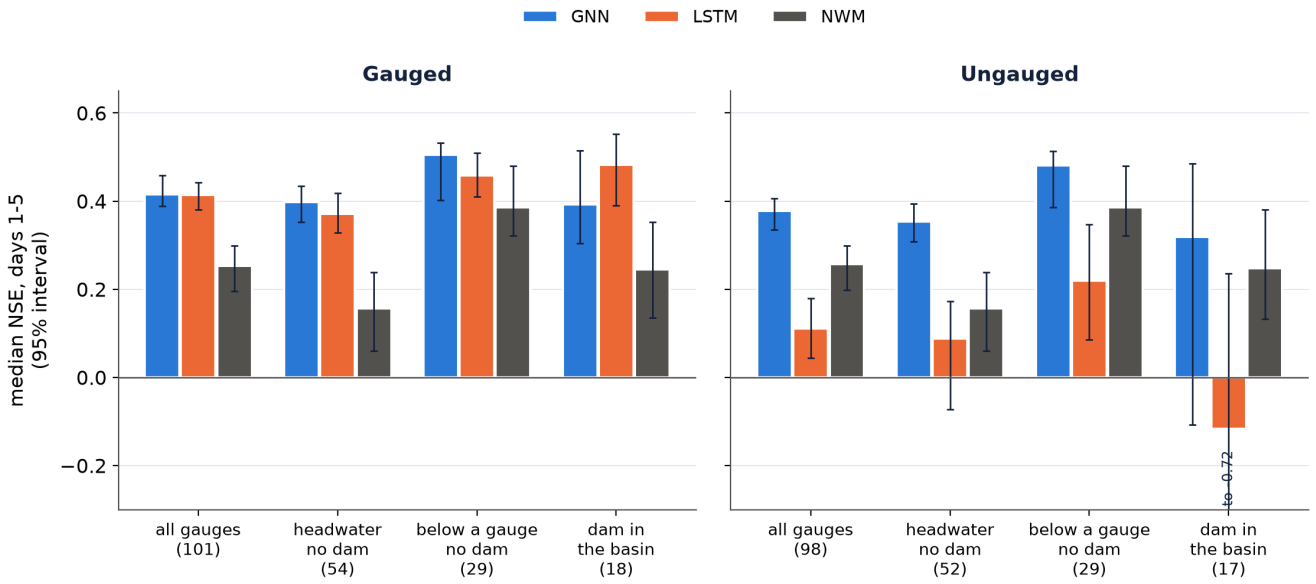


Figure 9. Days 1-5 median NSE by gauge type: headwater basins without a gauge or dam upstream, gauges below another gauge without a dam, and gauges with a TVA dam in the basin (95% intervals; the models never read the dam data).

Days 1-5, median NSE	Gauges	GNN	LSTM	NWM
Gauged: headwater, no dam	54	**0.399**	0.371	0.157
Gauged: below a gauge, no dam	29	**0.505**	0.459	0.386
Gauged: dam in the basin	18	0.392	**0.482**	0.245
Ungauged: headwater, no dam	52	**0.353**	0.088	0.157
Ungauged: below a gauge, no dam	29	**0.481**	0.220	0.386
Ungauged: dam in the basin	17	**0.318**	–0.117	0.247

At gauges with their flow shown, the GNN is ahead in headwater basins and below other gauges, and behind the LSTM where a dam is in the basin: reservoir releases are visible in the gauge's own history, which the LSTM exploits better. Without the gauge, the GNN is the best model in every group; the ungauged dam group has only 17 gauges and wide intervals.

5.2 By year and season

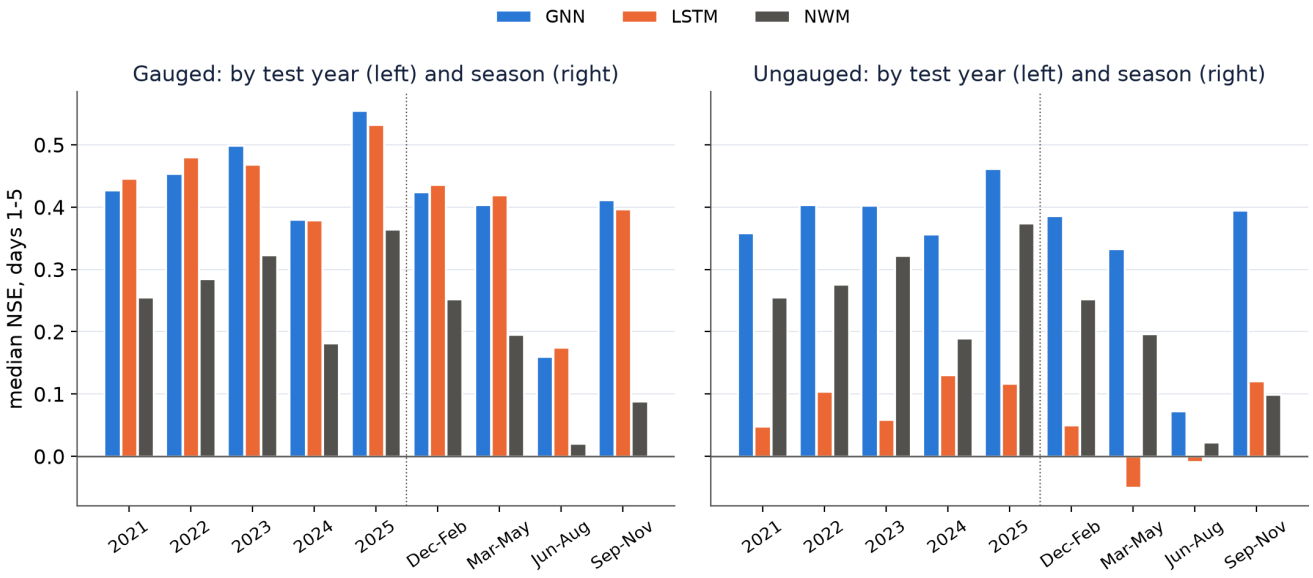


Figure 10. Days 1-5 median NSE using only the issues of each test year (2020 has only three months of test and is left out) and of each season.

The ranking is stable from year to year: gauged, the GNN and the LSTM trade places within a few hundredths; ungauged, the GNN leads every year and every season. Summer (June-August) is the hardest season for all models, when flow is low and much of the rain comes from convective storms, which are likely the hardest for the GEFS forecast to place.

5.3 Beyond NSE: correlation, variability and bias

Days 1-5, median over gauges	NSE	KGE	Correlation r	Variability ratio	Bias ratio
Gauged: GNN	0.415	0.518	0.661	0.690	0.985
Gauged: LSTM	0.414	0.552	0.656	0.725	0.996
Gauged: NWM	0.253	0.449	0.585	0.795	0.863
Ungauged: GNN	0.378	0.416	0.644	0.613	0.937
Ungauged: LSTM	0.111	0.073	0.602	0.490	0.853
Ungauged: NWM	0.258	0.456	0.587	0.796	0.866

KGE splits the error into timing (correlation), size of the swings (variability ratio, 1 is right) and volume (bias ratio, 1 is right). The GNN has the best timing and nearly unbiased volume, but its forecasts are too smooth (variability ratio 0.69 gauged, 0.61 ungauged). This is why NWM, whose swings are larger, has a slightly higher KGE at ungauged gauges even though its NSE is lower, and it is the same weakness that shows up in the flood peaks (section 7).

6. Every reach

No observations exist at 99.8% of reaches, so the forecasts are compared with the NWM v3.0 retrospective, NOAA's simulation driven by the weather that occurred. It is the best available reference, but not truth.

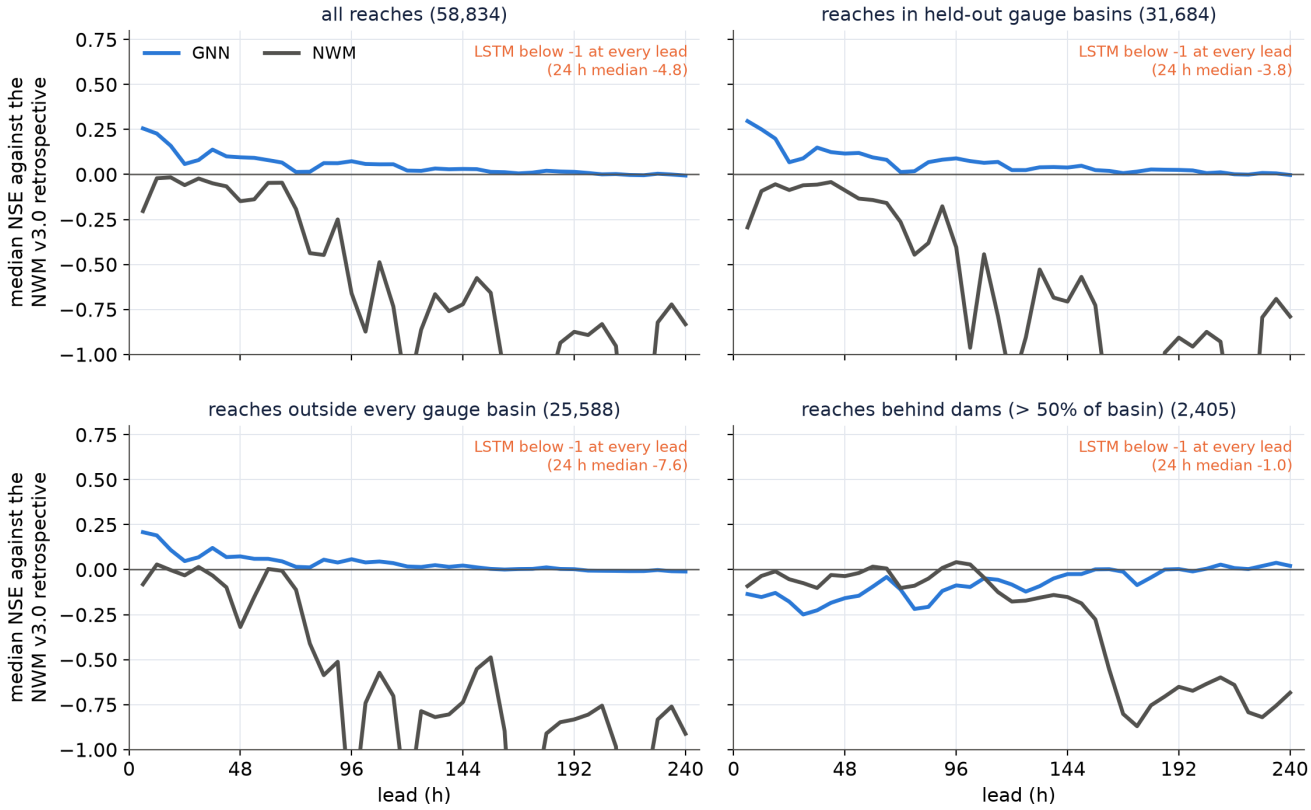


Figure 11. Median NSE against the NWM retrospective by lead, for all reaches, reaches in the basins of held-out gauges, reaches outside every gauge basin, and reaches with more than half of their basin behind a TVA dam.

Median NSE vs retrospective	Reaches	24 h: GNN	LSTM	NWM	72 h: GNN	LSTM	NWM
all reaches	58,834	**0.06**	–4.82	–0.06	**0.01**	–6.38	–0.19
reaches in held-out gauge basins	31,684	**0.07**	–3.77	–0.09	**0.01**	–5.02	–0.26
reaches outside every gauge basin	25,588	**0.05**	–7.58	–0.03	**0.02**	–9.40	–0.11
reaches behind dams (> 50% of basin)	2,405	–0.18	–1.02	**–0.05**	–0.11	–1.37	**–0.10**

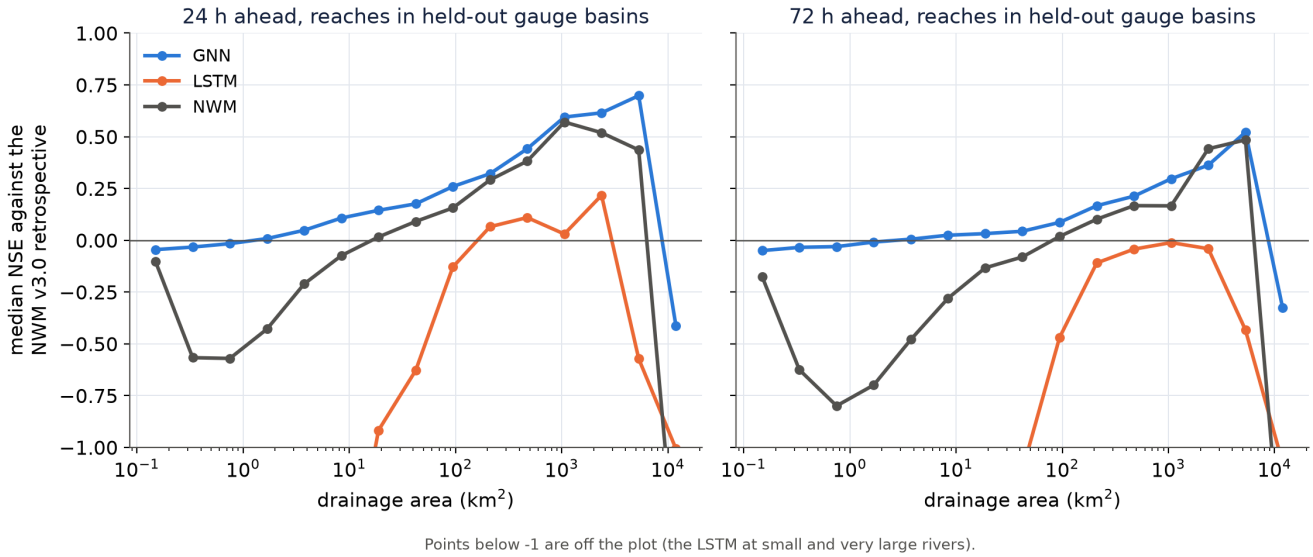


Figure 12. Median NSE against the retrospective by drainage area, reaches in held-out gauge basins.

Held-out basins, median NSE	Reaches	24 h: GNN	LSTM	NWM	72 h: GNN	LSTM	NWM
< 50 km^2	24,605	**0.04**	–7.38	–0.17	**0.00**	–9.74	–0.41
50-500 km^2	4,223	**0.28**	–0.04	0.24	**0.12**	–0.26	0.04
500-5,000 km^2	2,575	**0.60**	0.04	0.49	**0.34**	–0.07	0.33
> 5,000 km^2	281	**–0.31**	–1.02	–0.87	**–0.24**	–1.09	–1.36

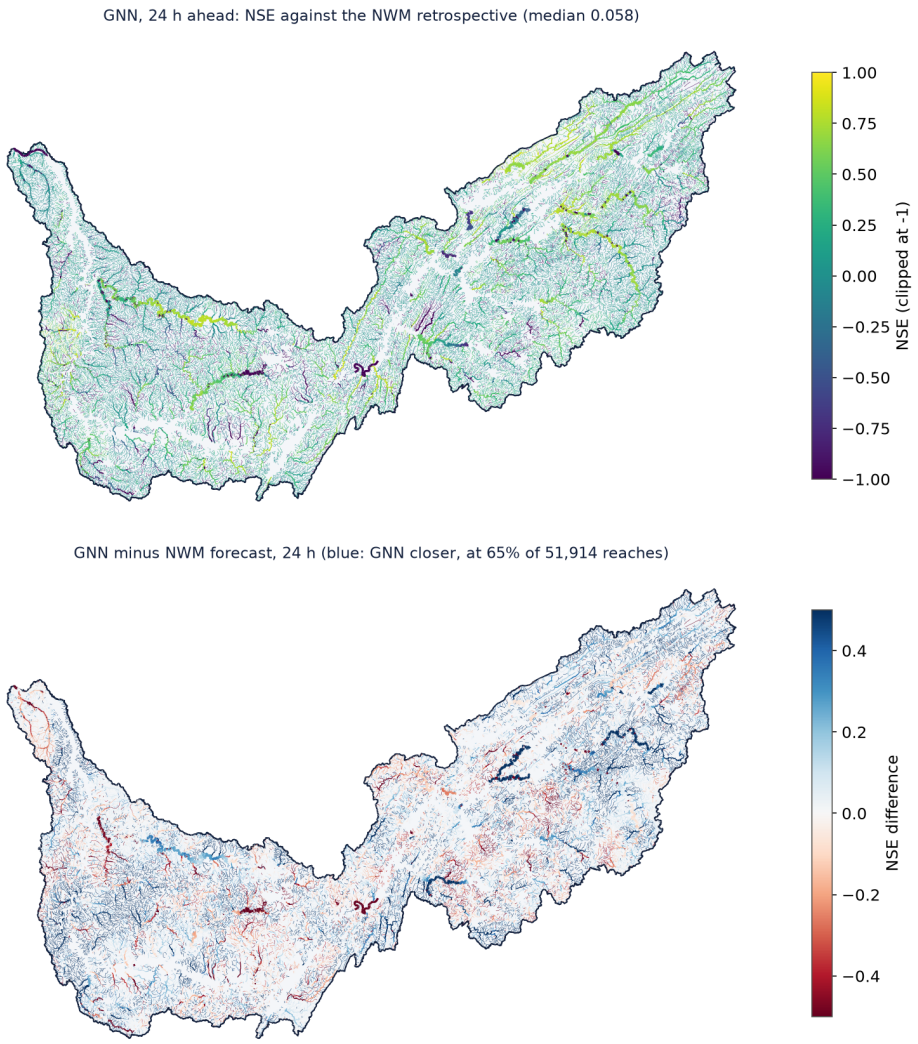


Figure 13. Every reach at 24 h. Top: the GNN's NSE against the retrospective. Bottom: the GNN's NSE minus the NWM forecast's (blue: the GNN is closer). Wider lines are larger rivers.

- The GNN is closer to the retrospective than the NWM forecast at 65% of reaches at 24 h and 70% at 72 h, and than the LSTM at 88%.
- It leads in every drainage-area class. Small streams (most of the network) have low scores for every model: their flow is flashy and the retrospective itself is uncertain there.
- The NWM forecast drifts away from its own retrospective after about three days (median below –1 at 120 h).
- Behind dams, the GNN is slightly behind NWM up to about 3 days; the main stem below the large reservoirs (> 5,000 km^2) is poor for every model, because flow there follows reservoir operations that no model is given.

7. Floods

7.1 Detecting high flows

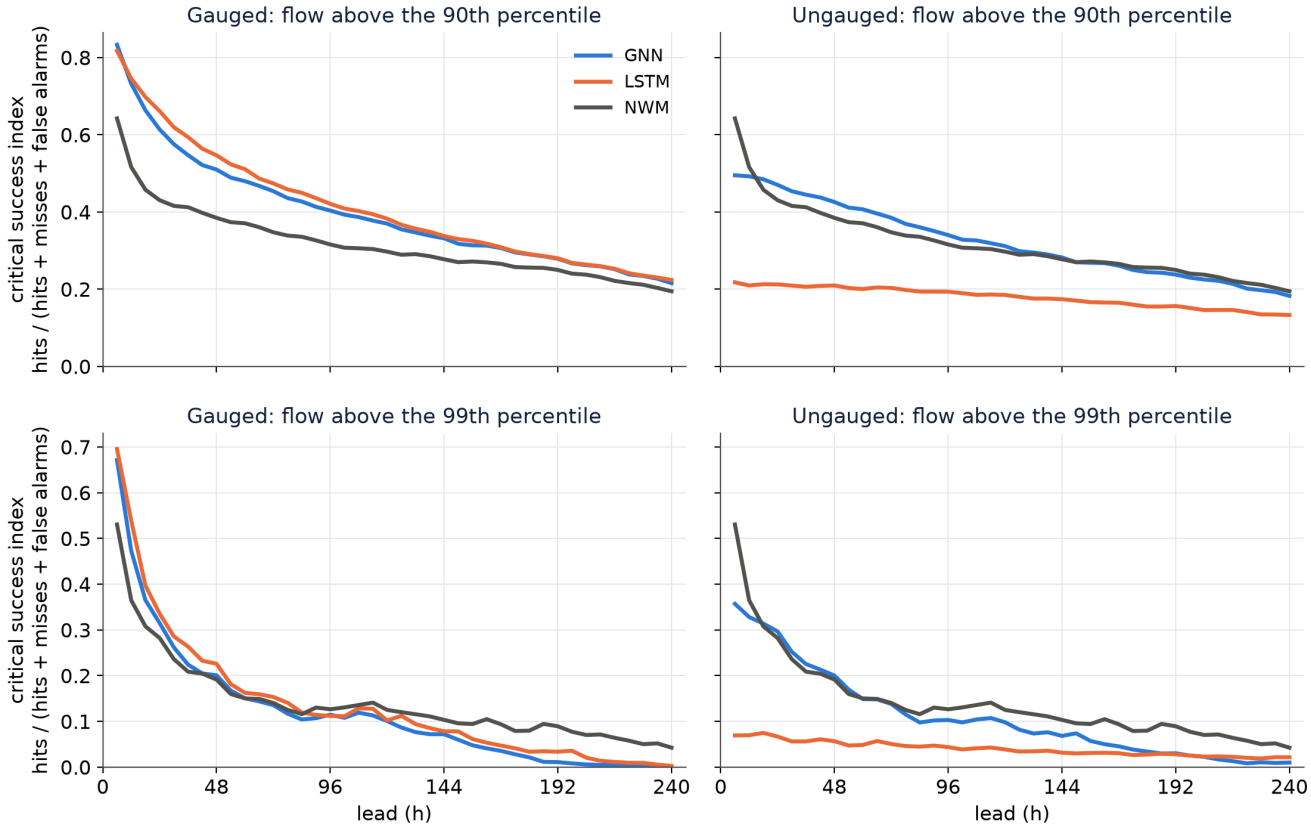


Figure 14. Critical success index for flows above each gauge's 90th (top) and 99th (bottom) training percentile: the share of high-flow forecasts or observations that were correct (1 is perfect).

Critical success index	24 h: GNN	LSTM	NWM	72 h: GNN	LSTM	NWM
Gauged, above 90th pct	0.614	**0.661**	0.430	0.454	**0.474**	0.347
Gauged, above 99th pct	0.315	**0.335**	0.282	0.135	**0.153**	0.140
Ungauged, above 90th pct	**0.470**	0.212	0.430	**0.385**	0.203	0.347
Ungauged, above 99th pct	**0.297**	0.067	0.282	0.138	0.050	**0.140**

At gauges, the LSTM detects high flows slightly better than the GNN in the first days, and both are ahead of NWM above the 90th percentile. Without the gauge, the GNN is slightly ahead of NWM and the LSTM falls far behind, mostly through false alarms (false-alarm ratio 0.73 at 24 h). Above the 99th percentile, skill is low for every model beyond 2 days.

7.2 The size of flood peaks

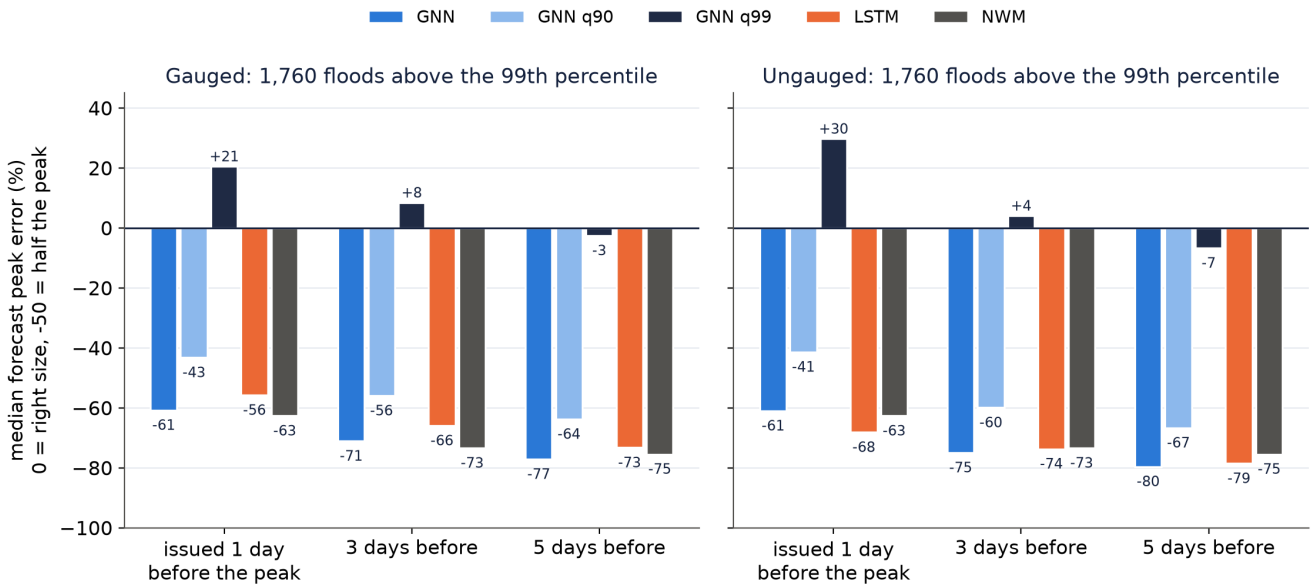


Figure 15. Median error in the size of observed peaks above the 99th percentile, for forecasts issued 1, 3 and 5 days before the peak (0 = right size). "GNN q90" and "GNN q99" are the 90th and 99th percentile forecasts of the GNN's tail head.

All three models underpredict large peaks by more than half, a known weakness of models trained on mean-squared error. The GNN's tail head offers a usable correction: its 99th-percentile forecast is close to the observed peak size (+21%, +8% and –3% at 1, 3 and 5 days, gauged). The quantiles are well calibrated at gauges whose flow is shown (share of observations below the 50th, 90th and 99th percentiles: 0.50, 0.91, 0.98) and somewhat too narrow at held-out gauges (0.46, 0.82, 0.96).

7.3 Examples

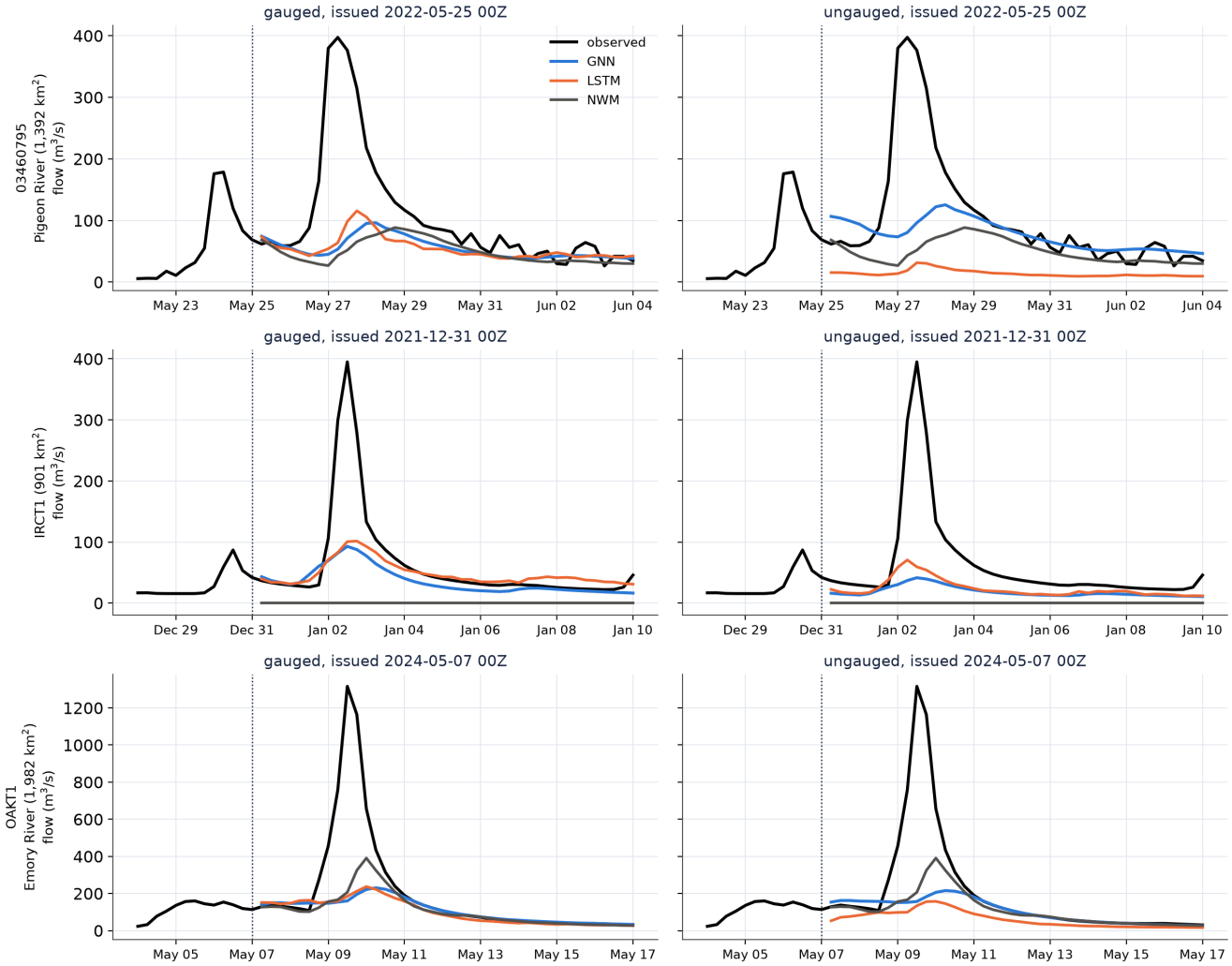


Figure 16. Three large floods (peaks 1.3-3 times the gauge's 99th percentile, at held-out gauges with basins over 300 km^2), forecast two days before the peak. Left: the gauge's recent flow is shown to the GNN and LSTM. Right: each model forecasts without ever having seen the gauge. Timing is mostly right; peak size is not. The NWM forecast is near zero at the IRCT1 reach. (The largest test floods outside Helene, flash floods in August 2021, were missed entirely by every model.)

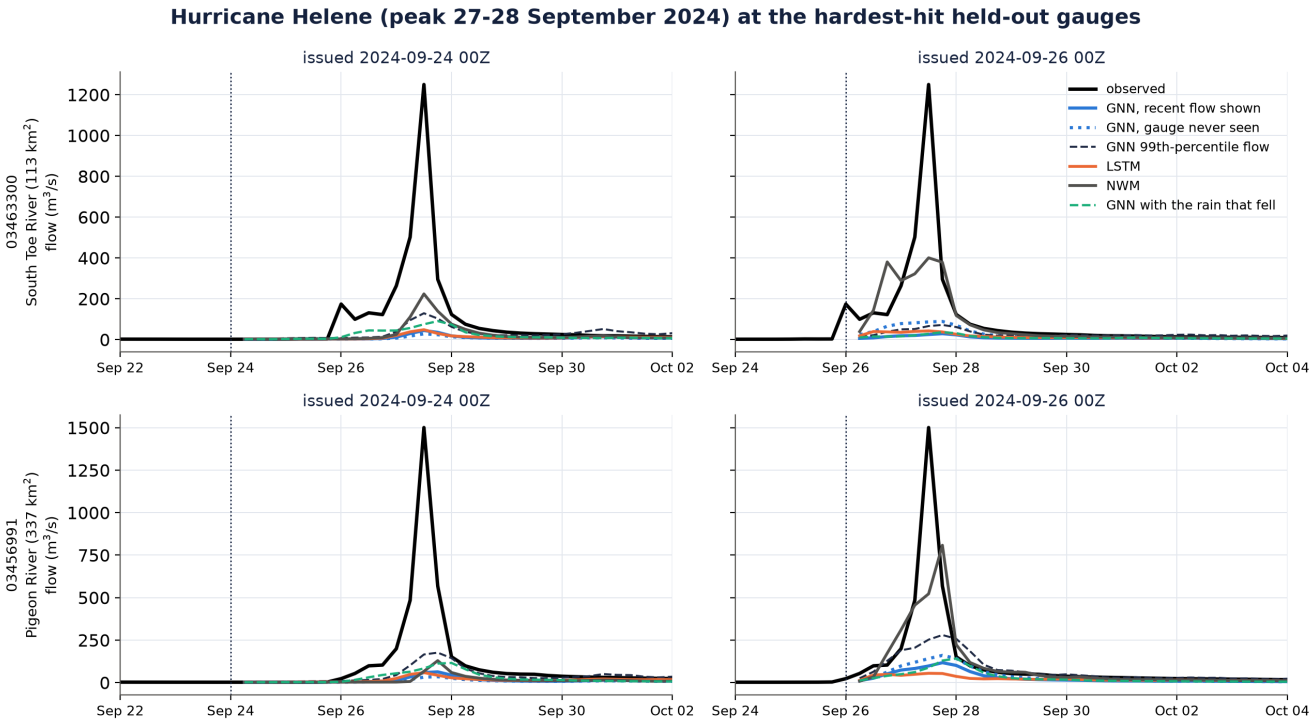


Figure 17. Hurricane Helene at the two hardest-hit held-out gauges, with peaks 28-45 times the gauge's 99th percentile. Every model is far too low; NWM, issued one day before, comes closest. Even with the rain that fell (green), the GNN stays low: the event is far beyond the flows it was trained on.

8. Headwater basins without dams

The Tennessee basin is heavily regulated: below the TVA dams, flow follows reservoir operations that no model here is given. To see how the models compare on natural flow, this section repeats the tests on **headwater gauges**: gauges with no TVA dam and no other gauge upstream (the "headwater, no dam" group of section 5.1). For the every-reach test it keeps the 55,738 reaches with no dam upstream (95% of the network; the dams' influence is concentrated on the main stems).

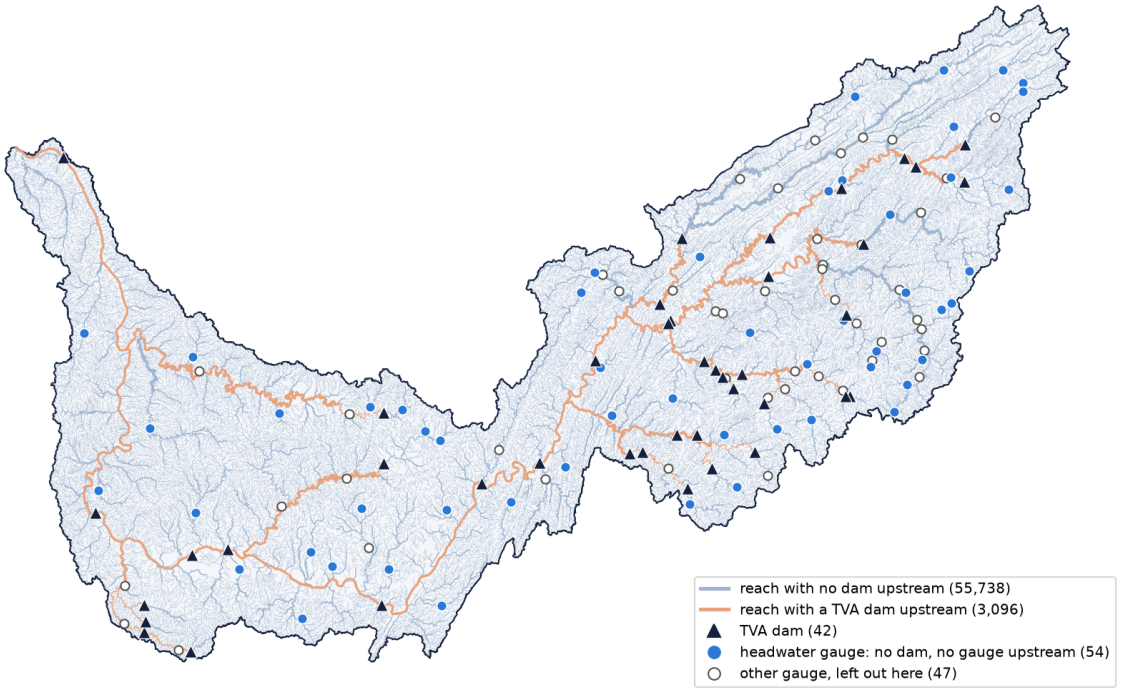


Figure 18. Where the 42 TVA dams reach. Orange: reaches with a dam upstream. Blue: the 54 headwater gauges used in this section. White: the other 47 scored gauges, left out here.

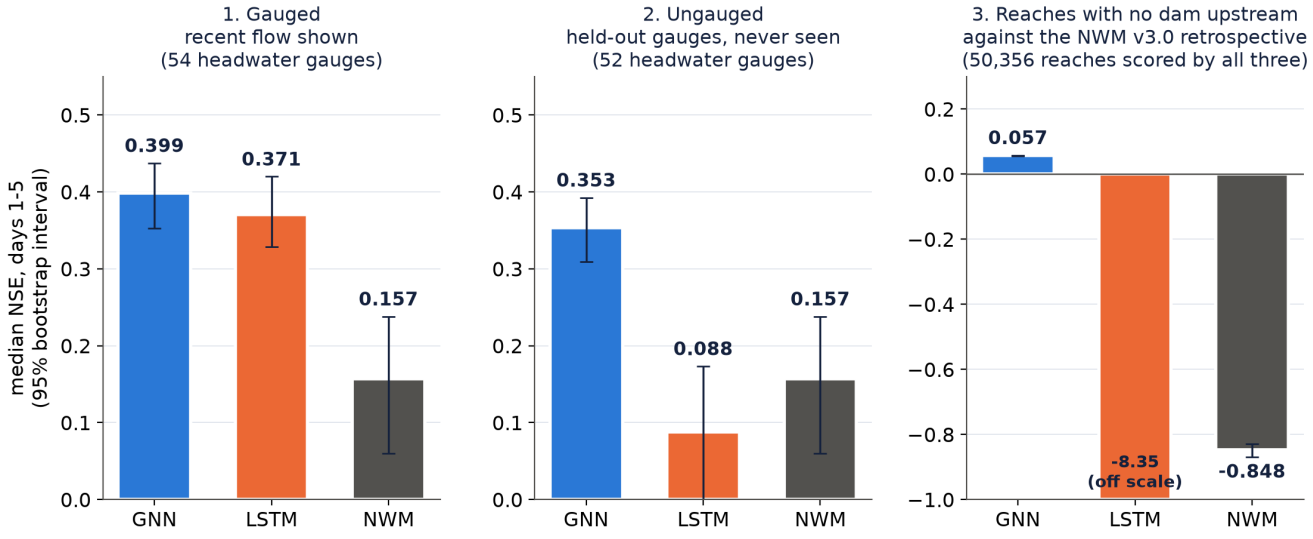


Figure 19. The three tests on headwater gauges and on reaches with no dam upstream: median NSE over days 1-5 with 95% intervals.

Headwater gauges, median NSE	6 h	24 h	72 h	120 h	240 h	Days 1-5
Gauged: GNN	0.915	0.549	0.357	0.310	0.145	0.399
Gauged: LSTM	0.916	0.550	0.343	0.254	0.074	0.371
Gauged: NWM forecast	0.816	0.347	0.180	–0.052	–0.449	0.157
Ungauged: GNN	0.731	0.523	0.322	0.211	0.094	0.353
Ungauged: LSTM	0.308	0.209	0.066	0.015	–0.046	0.088
Ungauged: NWM forecast	0.817	0.347	0.180	–0.052	–0.449	0.157

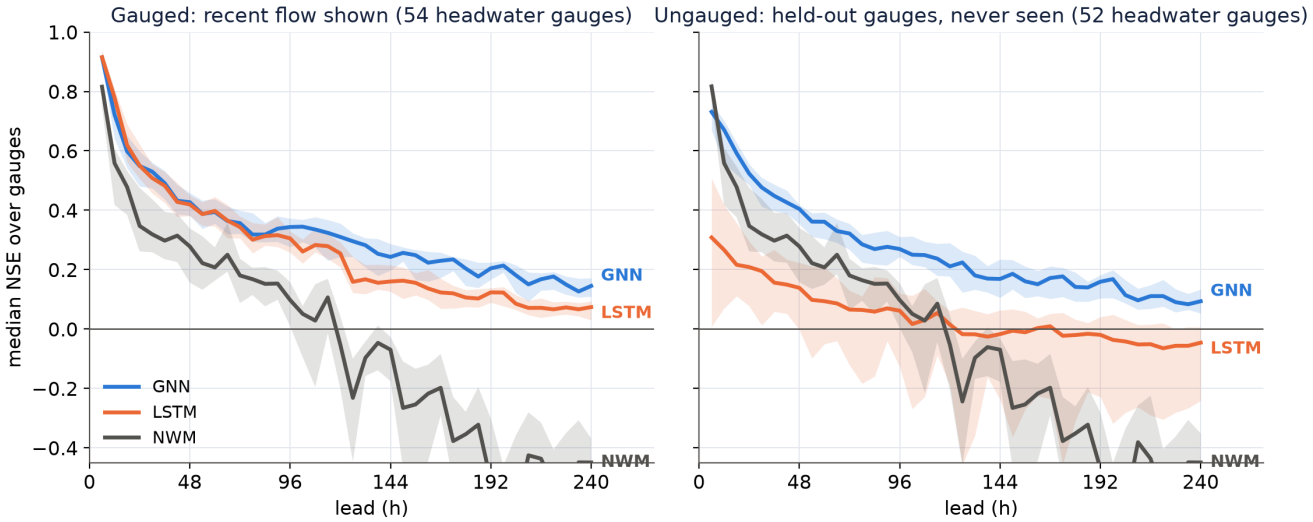


Figure 20. Median NSE by lead on headwater gauges (bands: 95% intervals).

- **Gauged:** without dam effects the GNN is ahead of the LSTM over days 1-5 (0.399 against 0.371), where on all gauges the two tie. They are level in the first day (24 h: 0.549 against 0.550) and the GNN leads from about 3 days (120 h: 0.310 against 0.254). The LSTM's edge at gauges in section 4 therefore comes from the gauges below dams. The GNN beats NWM at 94% of headwater gauges.
- **Ungauged:** 0.353 against 0.088 for the LSTM and 0.157 for NWM; the GNN is better than the LSTM at 90% and than NWM at 87% of held-out headwater gauges.
- **NWM is weaker in headwaters** (0.157 against 0.253 on all gauges), likely because small, flashy basins are the hardest for a physics model driven by forecast rain.

Headwater gauges, days 1-5	NSE	KGE	Correlation r	Variability ratio	Bias ratio
Gauged: GNN	0.399	0.500	0.638	0.639	0.976
Gauged: LSTM	0.371	0.502	0.636	0.710	0.999
Gauged: NWM	0.157	0.370	0.495	0.788	0.826
Ungauged: GNN	0.353	0.407	0.640	0.562	0.873
Ungauged: LSTM	0.088	0.057	0.585	0.669	0.976
Ungauged: NWM	0.157	0.382	0.495	0.792	0.829

Headwater gauges: floods	CSI 24 h: GNN	LSTM	NWM	Peak error, 1 day: GNN	LSTM	NWM
Gauged, above 90th pct	0.621	**0.668**	0.417			
Gauged, above 99th pct	0.286	**0.304**	0.268	–64%	–60%	–68%
Ungauged, above 90th pct	**0.483**	0.220	0.417			
Ungauged, above 99th pct	**0.305**	0.071	0.268	–66%	–65%	–68%

On KGE the GNN ties the LSTM at headwater gauges and is the best model without the gauge; its forecasts remain too smooth (variability ratio 0.64). Flood results match the whole basin: the LSTM detects high flows slightly better with the gauge shown, the GNN is best without it, and every model underpredicts peaks above the 99th percentile by about two thirds (1,062 headwater floods).

Reaches with no dam upstream	Reaches	24 h: GNN	LSTM	NWM	120 h: GNN	LSTM	NWM
all	55,738	**0.06**	–5.23	–0.06	**0.02**	–8.41	–1.29
< 50 km^2	45,703	**0.04**	–8.70	–0.13	**0.01**	–13.91	–1.64
50-500 km^2	7,749	**0.27**	–0.15	0.22	**0.14**	–0.47	–0.55
> 500 km^2	2,286	0.61	0.28	**0.63**	**0.28**	0.07	0.14

Across reaches with no dam upstream the GNN is closer to the retrospective than the NWM forecast at 66% of reaches at 24 h, 71% at 72 h and 89% at 120 h. On rivers over 500 km^2 NWM is slightly ahead at 24 h, level at 72 h, and behind from then on.

9. Against TVA's own forecasts

TVA forecasts flow at its headwater gauges twice a day, and its archive holds those forecasts as issued. We score TVA's 00 UTC forecasts on the same issues, gauges and observations as our models (a TVA value at lead L is the six-hour mean ending at issue + L, as for the models). 29 TVA gauges have TVA forecasts, observations and every model's forecast: 20 are headwater gauges without a dam and 9 lie below another gauge without a dam, so none is regulated. Every model is shown the gauge's recent flow, as TVA's forecast is. TVA has no ungauged forecast, so the GNN that never saw the gauge is shown only as a reference: our model without the gauge against TVA with it. We score TVA's forecasts as archived and do not know what inputs or manual adjustments went into them.

9.1 Skill by lead time

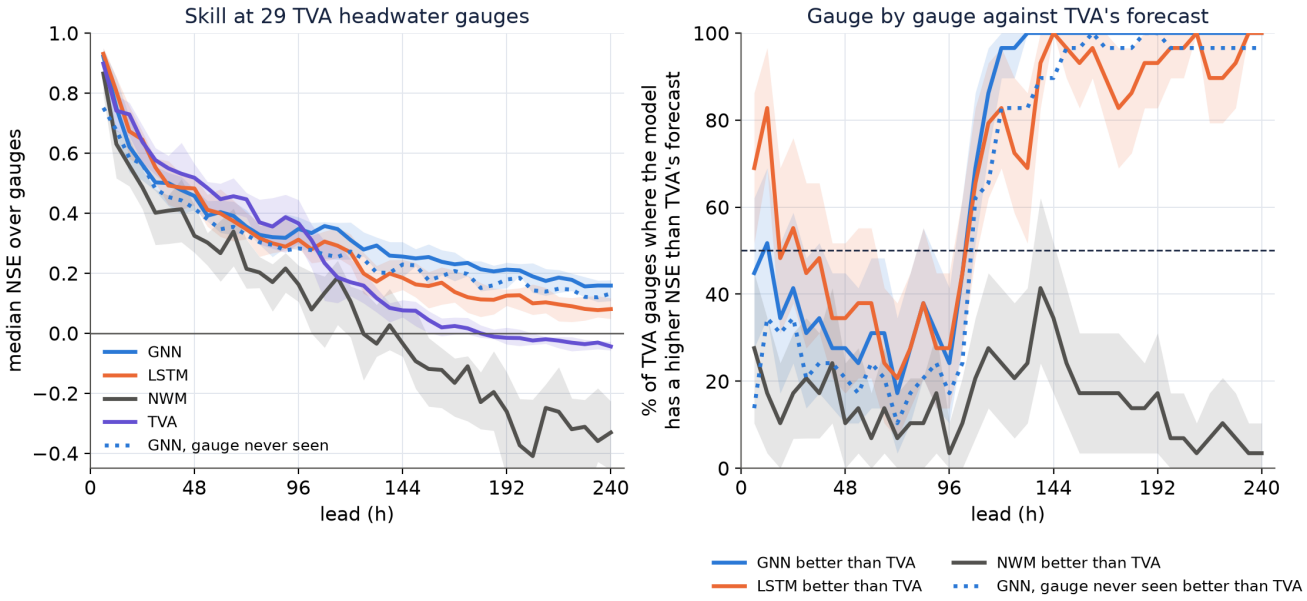


Figure 21. Left: median NSE by lead at the 29 TVA headwater gauges (bands: 95% intervals). Right: share of these gauges where each model has a higher NSE than TVA's forecast.

TVA gauges, median NSE	6 h	24 h	72 h	120 h	240 h	Days 1-5
GNN	0.922	0.561	0.355	0.310	0.158	0.398
LSTM	0.931	0.645	0.346	0.269	0.080	0.422
NWM forecast	0.865	0.486	0.214	0.106	–0.331	0.251
TVA forecast	0.898	0.637	0.446	0.170	–0.044	0.435
GNN, gauge never seen (reference)	0.751	0.563	0.327	0.272	0.132	0.386

- **Days 1-3: TVA is the best forecast.** At 72 h its median NSE is 0.446 against 0.355 (GNN) and 0.346 (LSTM), and it beats the GNN at 83% of the gauges.
- **Day 4 on: the GNN is the best forecast.** TVA's skill falls quickly after its rain forecast ends at about 72 h; at 120 h the GNN scores 0.310 against 0.170 and is better at 97% of gauges, and at 240 h at all of them. TVA drops below zero after about a week.
- **Over days 1-5 the three are close:** TVA 0.435, LSTM 0.422, GNN 0.398, with overlapping intervals. Without ever seeing the gauge, the GNN still reaches 0.386.

9.2 Floods

TVA gauges: floods	GNN	LSTM	NWM	TVA
CSI above 90th pct, 24h	0.633	**0.677**	0.432	0.604
CSI above 90th pct, 72h	0.469	**0.479**	0.362	0.471
CSI above 99th pct, 24h	0.284	0.306	0.288	**0.361**
CSI above 99th pct, 72h	0.111	0.130	0.122	**0.169**
Peak error, 1 day ahead (569 floods)	–63%	–58%	–68%	–50%
Peak error, 3 days ahead (569 floods)	–73%	–68%	–77%	–64%
Peak error, 5 days ahead (569 floods)	–78%	–74%	–79%	–92%

TVA gauges, days 1-5	NSE	KGE	Correlation r	Variability ratio	Bias ratio
GNN	0.398	0.510	0.638	0.660	0.985
LSTM	0.422	0.544	0.661	0.717	0.993
NWM	0.251	0.427	0.576	0.761	0.827
TVA forecast	0.435	0.530	0.681	0.754	0.900

For floods TVA is the strongest in the first days: it detects flows above the 99th percentile best at 24 and 72 h and gets closest to the peak size one and three days ahead (–50% and –64%, against –63% and –73% for the GNN). Five days ahead, beyond its rain forecast, TVA's peaks are the lowest (–92%). TVA's forecasts have the highest correlation with the observations (0.68) and are less smooth than the GNN's (variability ratio 0.75 against 0.66), but they underestimate volume by about a tenth (bias ratio 0.90).

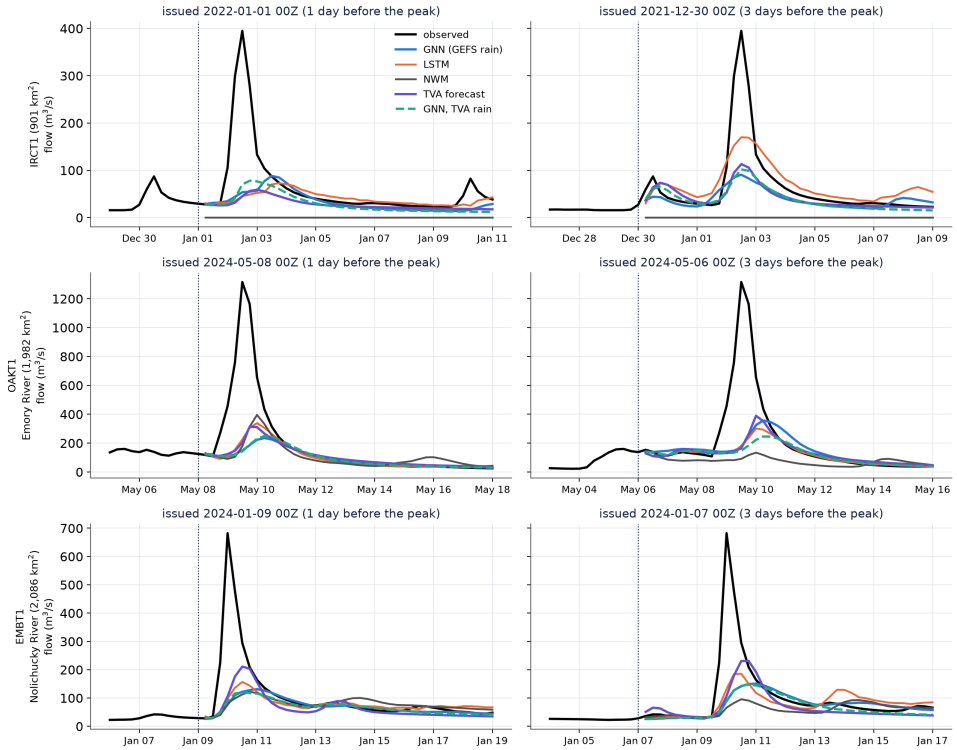


Figure 22. The three largest test floods at TVA gauges outside Hurricane Helene (peaks 1.3-3 times the gauge's 99th percentile, distinct gauges), issued one day (left) and three days (right) before the peak. Dashed green: the GNN with TVA's rain (section 9.3). The NWM forecast is near zero at the IRCT1 reach.

9.3 The GNN with TVA's own rain forecast

TVA's forecasts use TVA's own basin rain forecast, which is real to about 72 h and almost zero after about 5 days, while our models receive the GEFSv12 control member. To remove that difference we re-ran the GNN at every issue where TVA issued a forecast, replacing the rain over each TVA gauge's basin by TVA's basin-mean rain forecast (every other reach keeps GEFS). Two variants: TVA's rain at every lead, and TVA's rain to 72 h followed by GEFS. The model is not retrained, and TVA's rain is a single basin mean, much coarser than the gridded rain the GNN was trained on, so this tests the rain input only. Like TVA's forecast, the GNN is shown the gauge's recent flow (all-gauge model).

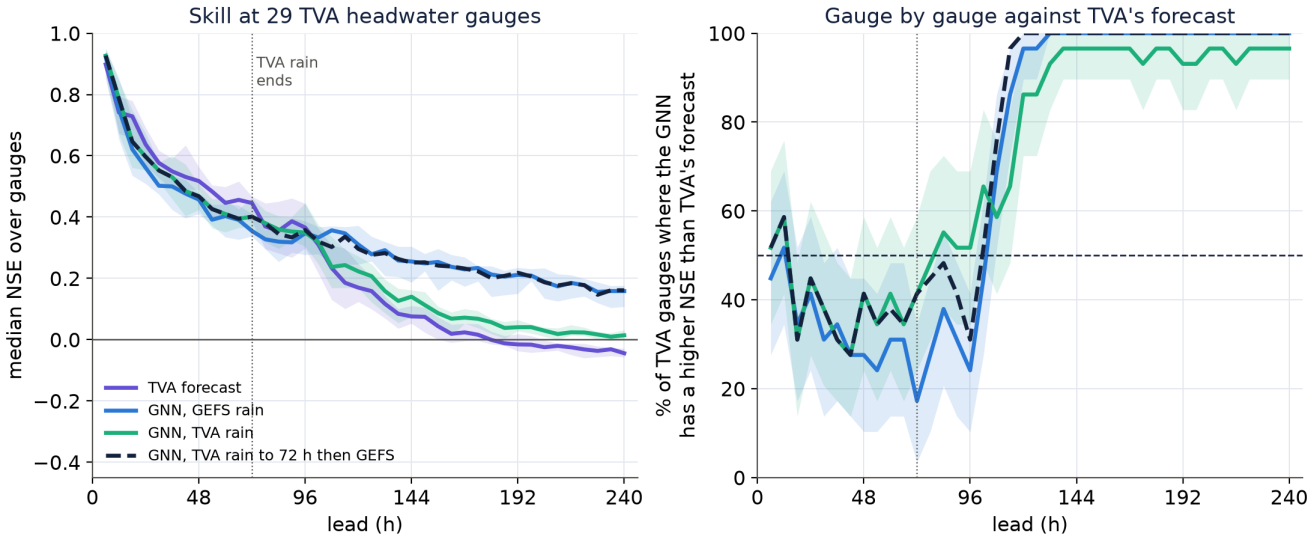


Figure 23. The GNN with GEFS rain, with TVA's rain, and with TVA's rain to 72 h then GEFS, against TVA's forecast at the same gauges. Dotted line: the end of TVA's rain forecast.

TVA gauges, median NSE	6 h	24 h	72 h	120 h	240 h	Days 1-5
TVA forecast	0.898	0.637	0.446	0.170	–0.044	0.435
GNN, GEFS rain	0.922	0.561	0.355	0.310	0.158	0.398
GNN, TVA rain	0.927	0.597	0.401	0.223	0.014	0.388
GNN, TVA rain to 72 h, then GEFS	0.927	0.597	0.401	0.296	0.161	0.400
GNN, rain that fell (diagnostic)	0.919	0.728	0.715	0.613	0.372	0.680

- **TVA's rain helps in days 1-3 but does not close the gap.** At 72 h the GNN goes from 0.355 to 0.401, about half of the way to TVA's 0.446; at 24 h from 0.561 to 0.597 (TVA 0.637). The GNN is better than TVA at 41% of gauges at 72 h, up from 17% with GEFS, and its one-day peak error shrinks from –63% to –52% (TVA –50%).
- **The rest of TVA's lead is not rain.** With the rain that fell the GNN reaches 0.715 at 72 h, so TVA's forecast rain is far from perfect, yet TVA's flow forecast still scores higher than the GNN driven by the same rain. TVA's model is calibrated to these basins, its forecasters can adjust it, and it may use inputs we do not have; the archive does not tell us which.
- **After TVA's rain ends the GNN needs GEFS.** With TVA's rain alone the GNN falls to 0.223 at 120 h; with GEFS after 72 h it keeps 0.296 (GEFS alone: 0.310) and beats TVA at 100% of gauges. This blend is the best GNN forecast at TVA gauges over days 1-5 (0.400, against 0.435 for TVA) and is what an operational version would use.

10. What the results mean

- **One model for gauged and ungauged use.** The GNN matches a strong, tuned LSTM at gauges and keeps almost all of that skill where there is no gauge, which neither baseline does. This is the main result.
- **The network matters most where there is no gauge.** Routing, graph layers and the NWM prior carry the forecast to reaches with no observations; the GNN is the only model with positive median skill across the network.
- **Long leads are a strength.** Beyond three days the GNN leads both baselines at gauges, and it degrades least out to 10 days.
- **Dams explain most of the LSTM's edge at gauges.** The LSTM uses a gauge's recent flow better in the first two days, mainly below dams, where that history encodes reservoir releases. In headwater basins without dams the GNN leads at gauges as well.
- **TVA's operational forecast is the strongest in the first three days** at its own gauges; the GNN is the strongest from day 4 on. Giving the GNN TVA's own rain forecast closes about half of the early gap.
- **Peaks are too low for every model.** The GNN's mean forecast is too smooth; its upper quantiles are the better basis for flood warnings. Floods far outside the training range (Helene) are missed by all models.

Caveats

- One training seed per model (GNN and LSTM); intervals cover the choice of gauges, not of seeds.
- NWM assimilates USGS observations and may use the held-out gauges, which favours it at short leads.
- The every-reach test measures agreement with the NWM retrospective, which the GNN is partly trained towards. It covers 425 issues to January 2023, when the retrospective ends.
- Rain is the GEFSv12 control member only; ensemble members and short-range rain (HRRR) are not used yet.
- The TVA comparison covers 29 gauges and uses TVA's forecasts as archived, with unknown inputs and forecaster adjustments. TVA has no ungauged forecast, so our ungauged results cannot be compared with TVA like for like.

11. Next steps

- Flood peaks: use the tail quantiles for warnings, and test combining the GNN with NWM at extreme flows, where NWM was closer during Helene.
- Regulated rivers: give the model what drives flow below dams (for example reservoir releases), the one place the LSTM still wins at gauges.
- Close the 1-3 day gap to TVA: TVA's rain forecast explains about half of it, so the rest needs short-range rain (HRRR) and a model trained on, or calibrated to, the rain products it will meet.
- A second seed of the GNN and the LSTM, to separate model differences from training noise.
- GEFS ensemble members, for probabilistic forecasts at every reach.

Glossary

Term	Meaning
Reach	A short NHDPlus river segment with its own small catchment; the basin has 58,834
Gauge	A reach with observed flow (USGS / CAMELS-H)
Issue time, lead time	When a forecast is made (00 UTC daily) and how far ahead a value is (6 h to 240 h)
Gauged / ungauged	Whether the model is shown the recent observed flow of the gauge it forecasts; ungauged gauges are forecast by a fold model that never saw them
Fold	One fifth of the gauges, withheld completely from one model for the ungauged test
NSE	Nash-Sutcliffe efficiency, 1 – squared error / variance of the observations: 1 is perfect, 0 is no better than the observed mean
KGE	Kling-Gupta efficiency, combining correlation, variability ratio and bias ratio; 1 is perfect
Days 1-5	Each gauge's mean NSE over the 24, 48, 72, 96 and 120 h leads; the median is then taken over gauges
CSI	Critical success index: hits / (hits + misses + false alarms) for flow above a threshold
NWM retrospective	NOAA's National Water Model v3.0 run with observed weather, 1979-2023: a simulation at every reach
GEFSv12	NOAA's Global Ensemble Forecast System; its control-member rain forecast is the only future input
Realized rain	The rain that actually fell: a perfect rain forecast, used only as a diagnostic
Persistence	Forecasting that the flow stays at its last observed value
TVA forecast	TVA's operational flow forecast at its headwater gauges, as archived (00 UTC issues used here)
Quantile (q90, q99)	A flow the observation should stay below 90% (99%) of the time

Where things are

- Code: `tva_flood_clean/src/tva_flood/hydro/`; training and evaluation `scripts/gnn_pipeline.sh`; method notes `docs/HYDRO.md`.
- Models: `/data/tva_data/models/gnn_B` and `gnn_B_f1..5` (GNN), `gefs_lstm*` (LSTM); evaluation `models/gnn_evaluation`.
- TVA comparison: TVA's archive `/data/tva_data/raw/tva/TVA_headwater.zip`; the GNN with TVA's rain `models/gnn_tva_rain`, made by `scripts/tva_rain_forecasts.py`.
- Figures and every number in this report: `/data/tva_data/figures/final_report/` (`tables.json`), made by `scripts/report_figures.py`.